# A candidate human surface-protein universe, *merged from seven independent sources.*

The Deliverome Project · A reference universe for transparent cross-source comparison · 2026-04-17 (revised 2026-04-18)

ABSTRACT

Selecting therapeutic-delivery targets on the human cell surface requires an unbiased starting list – one that captures every credible claim of surface expression without prejudging which evidence type is most trustworthy. We merge seven independent sources: UniProt, Gene Ontology, CSPA (mass spec), Human Protein Atlas v25 (immunofluorescence), SURFY and DeepTMHMM (sequence-based predictors), and JensenLab COMPARTMENTS (literature-driven meta-database). All entries are reconciled against current UniProt primary accessions; each source contributes one independent surface-evidence call. The resulting universe contains **6,518 proteins** with at least one source flagging surface evidence under its own rule – **45** supported by all seven, **49** by every source except DeepTMHMM (partial cohort in this milestone). Most proteins are supported by only one or two sources – an expected pattern we preserve deliberately for downstream prioritization.

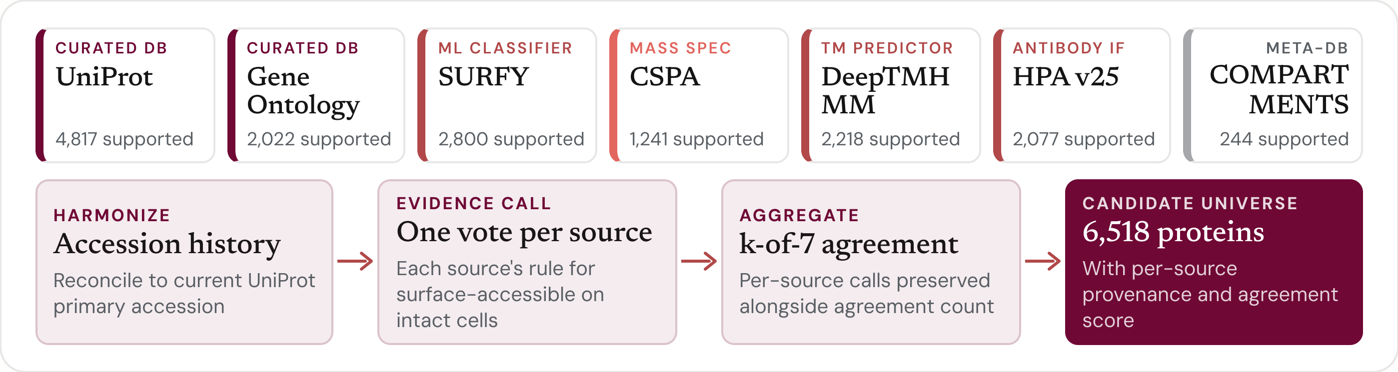



**FIGURE 1** The seven-source merge. Per-source calls aggregate into a *k*-of-7 agreement count carried downstream. Colors distinguish evidence modalities.

HEADLINE · 6,518 CANDIDATE PROTEINS

**45 supported by all 7 sources · 240 by at least 6**

*The tightest pairwise agreement is* **SURFY with DeepTMHMM** *(Jaccard 0.78) – both sequence-and-topology-driven. COMPARTMENTS contributes corroborating evidence only, never stand-alone admissions.*

## § 1 Data sources

| # | SOURCE | SNAPSHOT | EVIDENCE MODALITY | ROWS | SUPPORTED |
|---|---|---|---|---|---|
| 1 | UniProt human surface-candidate query (subcellular location + topology) | Release 2026_01 | Curated database | 4,817 | 4,817 |
| 2 | Gene Ontology annotations (three cellular-component roots — see § 2.2) | 2026-01 | Curated database | 2,557 | 2,022 |
| 3 | SURFY (Bausch-Fluck et al., 2018) | 2018 | ML classifier | 2,886 | 2,800 |
| 4 | CSPA (Bausch-Fluck et al., 2015) | 2015 | Mass-spec surfaceome | 1,500 | 1,241 |
| 5 | DeepTMHMM (canonical + isoforms) | Pre-existing cohort | TM topology predictor | 2,360 | 2,218 |
| 6 | Human Protein Atlas (subcellular localization) | v25 / 2026-01 | Antibody immunofluorescence | 3,004 | 2,077 |
| 7 | JensenLab COMPARTMENTS (four-channel integrated) | 2026-01 | Literature-driven meta-DB | 3,385 | 244 |

# § 2 Methods

*All seven sources are harmonized against UniProt release 2026_01*[3] *before the merge. For each source we describe what it measures, the rule under which a protein is counted, and the main reason that source disagrees with the others.*

## § 2.1 UniProt 4,817 SUPPORTED

**WHAT IT MEASURES** The Swiss-Prot reviewed human proteome[1], filtered to entries whose curator-assigned subcellular-location comments and topology features[2] place the protein on the plasma membrane.

**INCLUDED** Counted as UniProt-supported if any of six query clauses matches – subcellular-location comment contains `"Cell membrane"`, `"Cell surface"`, `"Apical cell membrane"`, `"Basolateral cell membrane"`, or `"GPI-anchor"` – or the entry carries a topological-domain feature annotated `Extracellular`. The disjunction captures single- and multi-pass TM proteins with an extracellular topology as well as GPI-/lipid-anchored proteins that lack a conventional TM helix.

**EXCLUDED** Unreviewed (TrEMBL) entries are excluded by design. Reviewed entries matching none of the six clauses above, and entries whose subcellular location is exclusively internal-organelle (ER, Golgi, lysosome, mitochondrion, nucleus) without a cell-surface comment or extracellular topology, do not enter the pool.

**WHY IT DISAGREES** Evidence depth varies by gene: well-studied proteins carry direct experimental annotations; less-studied ones rely on orthologue transfer. The `Extracellular` clause also admits some multi-location proteins; we retain them deliberately.

**QUERY DISJUNCTION**
Six clauses OR-ed: five subcellular-location keywords (`cc_scl_term`) plus the topology feature (`ft_topo_dom:Extracellular`).

**2026_01**
UNIPROT RELEASE

## § 2.2 Gene Ontology 2,022 SUPPORTED

**WHAT IT MEASURES** Curated gene–term associations for subcellular location[4]. Each association carries an evidence code – experimental, curated, sequence-inferred, or purely electronic.[5,6]

**INCLUDED** Three cellular-component roots plus their `is_a` + `part_of` descendants: **cell surface** (`GO:0009986`), **external side of plasma membrane** (`GO:0009897`), and **integral component of plasma membrane** (`GO:0005887`). Counted as GO-supported if the protein has at least one association to an included root (or descendant) backed by an experimental, curated, or non-electronic sequence-inference code.

**EXCLUDED** The parent **plasma membrane** (`GO:0005886`) is excluded as too broad – it admits cytosolic-facing PM-associated proteins, and its PM-specific members are already covered by the children. The molecular-function terms **signaling receptor activity** (`GO:0038023`) and **TM signaling receptor activity** (`GO:0004888`) are excluded because they capture *activity* regardless of location, and admit endosomal pattern-recognition receptors (TLR3/7/8). The anchored-component terms (`GO:0031225`, `GO:0046658`) are obsolete in current GO; GPI/lipid- anchor evidence is captured via UniProt `ft_lipid`. Purely electronic (IEA-only) annotations are retained as provenance but do not count.

**WHY IT DISAGREES** GO coverage is deep for well-studied receptors but thinner for less-curated surface proteins; IEA-only support is excluded.

**WHY DROP MF TERMS**
Admitting `GO:0038023` / `GO:0004888` pulls in endosomal TLR3/7/8 and other intracellular-membrane signaling receptors on activity annotation alone – incompatible with "accessible on intact cells."

**IEA-ONLY**
Retained as provenance (`go_low_confidence_only = 1`); never counted.

# § 2 Methods *continued – SURFY & CSPA*

## § 2.3 SURFY 2,800 SUPPORTED

**WHAT IT MEASURES** A random-forest classifier trained by Bausch-Fluck et al.[7] on a curated gold-standard surfaceome using sequence, topology, and annotation features. Assigns every human protein a surface / not-surface label and a confidence score.

**INCLUDED** Counted as SURFY-supported if the protein's published SURFY label is `"surface"`. We use the label as provided by the authors; no additional score cutoff is re-applied.

**EXCLUDED** Proteins labeled `"nonsurface"` and entries left unclassified in the SurfaceomeMasterTable are excluded and dropped from the merge input (rather than carried as provenance) because SURFY covers the full reviewed proteome – keeping them would only inflate the zero-support bucket without adding information.

**WHY IT DISAGREES** SURFY tends to over-call. A protein with a transmembrane domain and sequence hallmarks of membrane insertion tends to score positive whether or not it actually reaches the plasma membrane.

> **KNOWN FAILURE**
> *ABCB9*, a lysosomal ABC transporter, is flagged by SURFY despite residing on an internal membrane. Strong disagreement between SURFY and the experimental sources (CSPA, HPA) is itself an informative signal.

## § 2.4 CSPA 1,241 SUPPORTED

**WHAT IT MEASURES** The Cell Surface Protein Atlas (Bausch-Fluck et al., 2015)[8]: a mass-spectrometry surfaceome built by chemically labeling N-linked glycopeptides on the surface of intact, living cells across many human cell lines. Each detection is categorized in CSPA Table_B as **1 – high confidence**, **2 – putative**, **3 – unspecific**, or blank (detected in Table_A but not assigned a Table_B class).

**INCLUDED** Counted as CSPA-supported if any pre-accession-history-collapse row has category **"1 – high confidence"** or **"2 – putative"**. The flag is a boolean OR across pre-collapse rows, so a primary that inherited evidence from multiple merged historical accessions (e.g. HLA alleles) gains the flag if any contributing row was high-confidence or putative.

**EXCLUDED** **"3 – unspecific"** detections (non-specific captures / contaminants) and **blank-category** rows (detected but not classified) are retained in the merge as provenance (`cspa_surface_flag = 0`) so downstream can distinguish "absent from CSPA" from "seen only as unspecific" from "detected but not classified." They do not contribute to the positive flag.

**WHY IT DISAGREES** CSPA only observes proteins expressed on the specific cell lines sampled. Absence is not evidence of non-surface status – only absence from the sampled contexts.

> **SCOPE CAVEAT**
> Low pairwise agreement between CSPA and the database-driven sources (0.08–0.32) reflects this sampling scope, not data quality.

# § 2 Methods *continued – DeepTMHMM & Human Protein Atlas*

## § 2.5 DeepTMHMM 2,218 SUPPORTED

**WHAT IT MEASURES** A deep-learning transmembrane-topology predictor[9]. Assigns one of five labels per sequence: `TM` (alpha-helical transmembrane), `SP+TM` (signal peptide + TM helices), `BETA` (transmembrane beta-barrel), `SP` (signal peptide only – predicted secreted), or `GLOB` (globular, no TM or SP).

**INCLUDED** Counted as DeepTMHMM-supported if any run on the canonical sequence or isoforms returns `TM` or `SP+TM` – the labels consistent with an alpha-helical plasma-membrane protein.

**EXCLUDED** `BETA` is excluded because in humans, beta-barrel membrane proteins are essentially mitochondrial outer membrane (VDAC1/2/3, TOMM40, SAMM50, MTX1/2) – topology is informative about fold but misleading about compartment. `SP` is excluded because it identifies secreted proteins, which are not surface-accessible on intact cells by definition. `GLOB` is excluded because it indicates no membrane segment. The raw label is preserved in the emitted table so downstream can re-inspect the exclusion reason.

**WHY IT DISAGREES** DeepTMHMM is run on a partial cohort of 2,360 proteins in this milestone – a scope limit, not a correctness issue.

**WHY DROP BETA**
In humans, beta-barrel predictions overwhelmingly identify mitochondrial outer membrane proteins (VDAC1/2/3, TOMM40, SAMM50) – not plasma-membrane / cell-surface proteins.

**SCOPE**
Extending DeepTMHMM to the full reviewed proteome is a planned next step; until then, its pairwise overlaps with the other sources are mechanically capped.

## § 2.6 Human Protein Atlas v25 2,077 SUPPORTED

**WHAT IT MEASURES** An antibody-based immunofluorescence atlas of subcellular localization.[10,11] Each gene is scored into named localizations (plasma membrane, cell junctions, vesicles, endosomes, lysosomes, Golgi, ER, nucleus, cytosol, mitochondrion, …) with a per-localization reliability tier: **Enhanced**, **Supported**, **Approved**, or **Uncertain**.[12]

**INCLUDED** Counted as HPA-supported if either **Plasma membrane** or **Cell Junctions** appears at **Enhanced**, **Supported**, or **Approved** in the per-localization tier column (not gene-wide Reliability). Cell junctions count: E-cadherin, claudins, JAM-family, occludin, and desmosomal cadherins carry genuine extracellular-accessible domains on intact epithelia.

**EXCLUDED** PM or Cell Junctions at **Uncertain** reliability are excluded. HPA's **Extracellular-location** column is excluded entirely: in v25 it is populated by "Predicted to be secreted" – a SignalP-based sequence prediction, not an imaging observation – so counting it would double-count a predictor signal and conflate secreted with surface-accessible. **Vesicle, endosome, and lysosome** staining is excluded from the flag because cycling between internal compartments and the surface is not evidence of steady-state surface residence; it is carried as provenance (`hpa_trafficking_associated`). Nuclear, cytosolic, Golgi, ER, and mitochondrial localizations are excluded as not surface-relevant.

**WHY IT DISAGREES** HPA coverage is antibody-dependent. Proteins without a validated antibody at the v25 release, or whose surface localization is limited to cell types not sampled, will be absent.

**WHY PER-TIER**
Using the *per-localization* tier (not the gene-wide Reliability) avoids overcalls where a gene's strongest signal is nuclear/cytosolic while its PM call lands in Uncertain.

**TRAFFICKING ≠ SURFACE**
Vesicle/endosome/lysosome staining carried as provenance only. Admitting it on its own would reintroduce ABCB9-class internal-membrane false positives.

# § 2 Methods *continued – JensenLab COMPARTMENTS*

## § 2.7 JensenLab COMPARTMENTS 244 SUPPORTED · CORROBORATION-GATED

WHAT IT MEASURES A meta-database (Binder et al., 2014)[13] integrating four evidence streams per (protein, location): a **knowledge** channel from curator-assigned annotations, an **experiments** channel primarily ingesting immunofluorescence, a **text-mining** channel from automated co-mention in biomedical literature, and a **predictions** channel from sequence-based localization predictors (WoLF PSORT + YLoc-HighRes). Scores are reported on a 0–5 star scale per channel per term.

INCLUDED The **text-mining** channel and the non-HPA portion of the **experiments** channel are used as independent evidence[14]. Counted as COMPARTMENTS-supported when both hold: (a) `max(experiments, textmining) ≥ 3` stars across the surface GO-term set `{GO:0005886, GO:0009986, GO:0031225, GO:0005887}`, and (b) at least one of the other six sources independently flags the protein as surface under its own rule (`compartments_corroborated == 1`).

EXCLUDED The **knowledge** channel is excluded because it re-ingests GO and UniProt-SubCell – both already first-class sources here; counting it would triple-count curator evidence. The **predictions** channel is excluded because it wraps WoLF PSORT + YLoc-HighRes – sequence-based predictors in the same family as SURFY and DeepTMHMM; empirically it drives ~73% of pre-filter hits and is ~100% PSORT-on-`GO:0005886`. Within the experiments channel, rows whose upstream source is "HPA" are dropped before scoring to avoid double-counting the first-class HPA IF evidence. All four channels' per-term stars are retained as provenance columns.

WHY IT DISAGREES By construction, COMPARTMENTS provides supporting evidence only; it cannot admit a protein on its own. Pairwise overlap with every other source is therefore low, and its own count (244) is small.

WHY CORROBORATION

Text-mining alone picks up legitimate co-occurrences with "plasma membrane" for transcription factors (TP53, MYC), cytokines (IL1B, IFNG, IL4), and secreted serum proteins (ALB, APOE, CRP) that are not surface-accessible on intact cells.

# *§ 2.8* Source dependencies

*The seven primary sources do not sit side by side – several ingest each other, and a handful of external upstream tools feed specific primaries. These edges matter when reading the k-of-7 agreement on the next page: apparently independent sources often share upstream inputs, and some sources (notably COMPARTMENTS) integrate evidence from several others.*

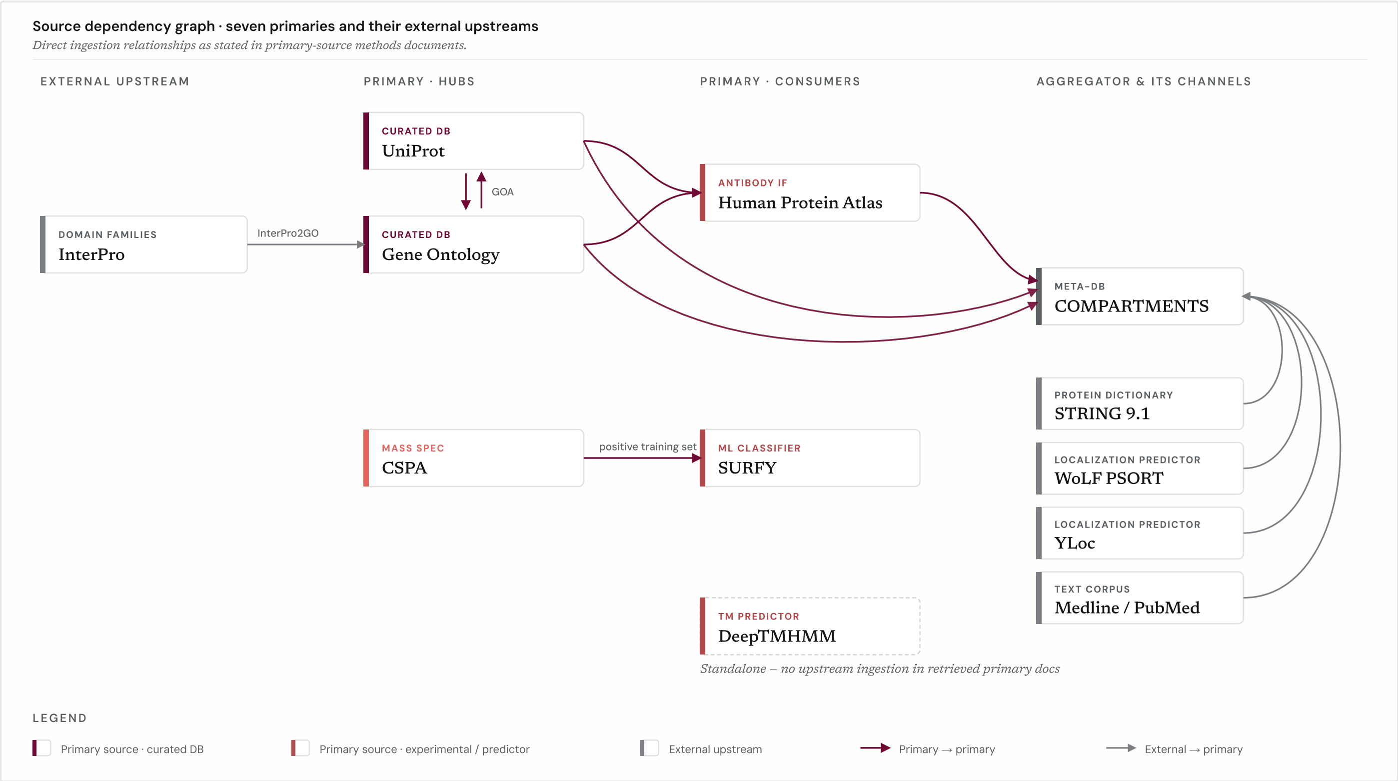
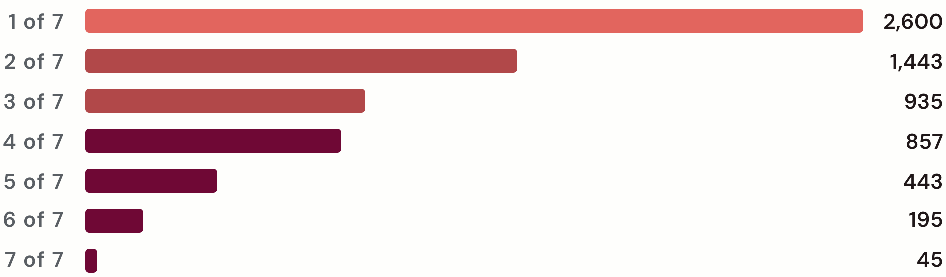



**FIGURE 2** **Source dependency graph.** Direct ingestion relationships as stated in the primary-source methods documents linked in §§ 2.1–2.7. UniProt ↔ Gene Ontology form a bidirectional hub via the GOA pipeline (*Keyword2GO*, *SubcellularLocation2GO*, *InterPro2GO*). CSPA feeds SURFY's positive training set. HPA consults UniProt for knowledge-based annotation and emits GO cellular-component identifiers. COMPARTMENTS is the main aggregation sink – integrating UniProt comments, GO term mapping with ontology propagation, HPA imaging data, STRING 9.1's protein dictionary, WoLF PSORT and YLoc predictions, and Medline abstracts. DeepTMHMM is standalone in the retrieved primary-source documentation.

# *§ 3* Results

*The agreement distribution and pairwise Jaccard overlaps quantify where the seven sources converge, and – more usefully – where they don't. Differences across sources are expected and help define where the evidence is strongest or still uncertain.*

## § 3.1 k-of-7 agreement distribution




Forty-five proteins are supported by all seven sources; another **49** by every source except DeepTMHMM (partial cohort). The 1-of-7 tail is dominated by proteins that only the UniProt curator-assigned query catches. A further 2,252 accessions appear in a raw source but fail every positive-flag rule (GO-IEA-only, CSPA unspecific/blank, HPA secreted-only, split-ambiguous remaps, COMPARTMENTS below corroboration); those are emitted to `candidate_universe_zero_support.tsv` and are not part of the main universe.

The 7-of-7 and 6-of-7 classes define a high-confidence core. The middle classes (2-of-7 through 5-of-7) are often the most biologically interesting – disagreement frequently carries real meaning (predictor over-calls, cell-line sampling gaps, missing HPA antibodies).

45
SUPPORTED BY ALL 7

240
SUPPORTED BY ≥ 6

100 %
IN DATABASE-SOURCE UNION

**INVARIANT**

By construction, every COMPARTMENTS-supported protein in the universe is also independently supported by another source. No proteins enter on literature-mined evidence alone.

## § 3.2 Pairwise agreement

| | |
|---|---|
| SURFY ∩ DeepTMHMM sequence / topology pair | 0.779 |
| UniProt ∩ SURFY | 0.520 |
| UniProt ∩ DeepTMHMM | 0.402 |
| CSPA ∩ DeepTMHMM | 0.324 |
| UniProt ∩ Gene Ontology | 0.310 |
| SURFY ∩ CSPA | 0.287 |
| Gene Ontology ∩ DeepTMHMM | 0.271 |
| Gene Ontology ∩ SURFY | 0.261 |
| UniProt ∩ HPA | 0.196 |
| HPA ∩ DeepTMHMM | 0.169 |
| CSPA ∩ HPA | 0.118 |
| UniProt ∩ COMPARTMENTS | 0.041 |

SURFY ∩ DeepTMHMM is expected: both derive calls from sequence and topology. UniProt pairs tightest with SURFY (0.52). GO's overlaps (0.26–0.31) reflect its cellular-component-only scope after MF receptor-activity terms were excluded. Low overlaps involving CSPA / COMPARTMENTS reflect sampling scope and the corroboration requirement, not data quality.

**WELL-KNOWN EXAMPLES**

EGFR is supported by all seven sources. CD20 (MS4A1) and HLA-A by six of seven. HLA-B (5/7), HLA-C (5/7), HLA-DQA1 (4/7), HLA-DRB1 (4/7), and IGHM (4/7) fall to four or five of seven – driven by both HPA's coverage gaps on individual HLA alleles and immunoglobulin class-switch variants *and* DeepTMHMM's partial-cohort scope in this milestone.

## § 3.3 Source-agreement figures

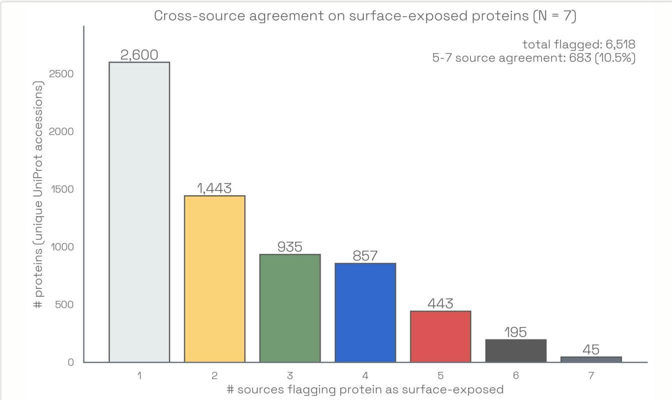



**FIGURE 3 k-of-7 agreement.** How many of the seven sources support each of the 6,518 proteins in the universe (`n_sources_surface ≥ 1`). 45 are supported by all seven; 240 by at least six. Most proteins sit in the lower agreement classes, reflecting genuine differences in what each source can see.

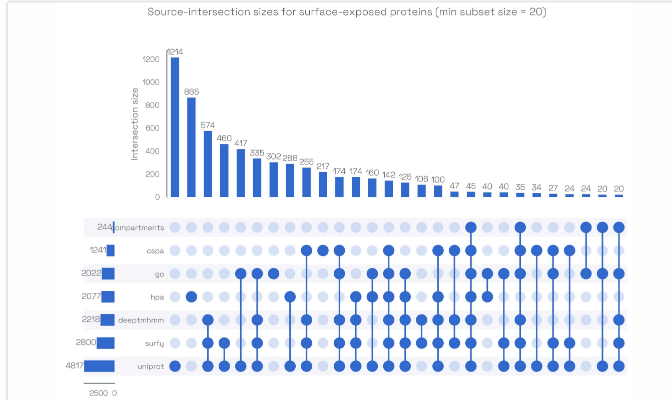



**FIGURE 4 Co-occurrence across sources.** The seven-way intersection (45 proteins) and the database-only intersection without DeepTMHMM (49 proteins) dominate the high-confidence end; intersections that require DeepTMHMM are capped by its partial coverage in this milestone.

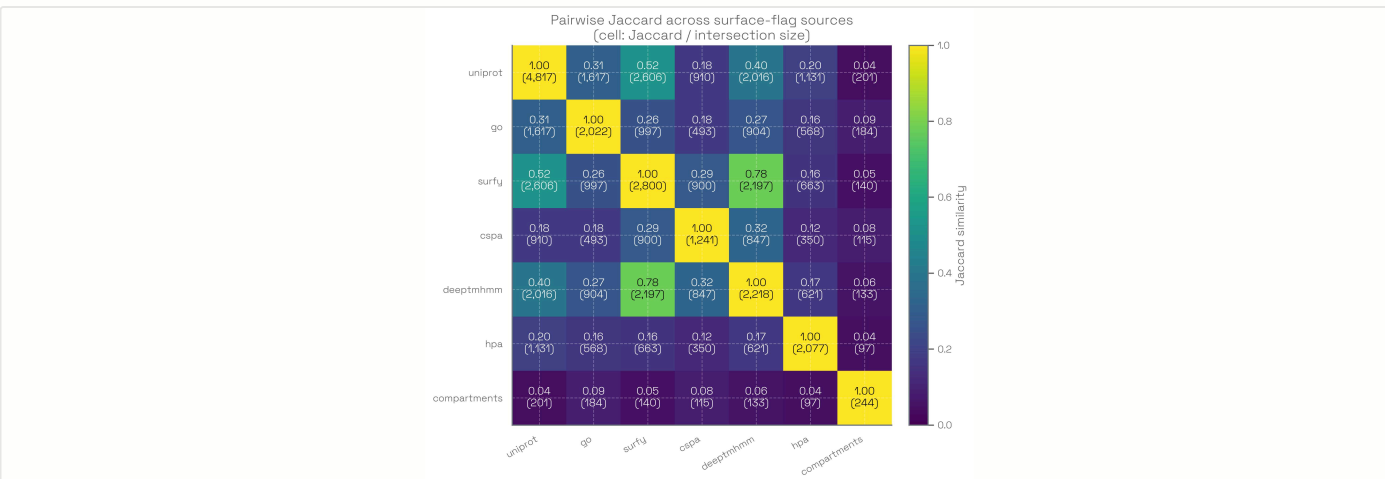



**FIGURE 5 Pairwise agreement across all seven sources.** High values along the SURFY–DeepTMHMM axis reflect their shared sequence-and-topology basis. Low values involving CSPA and COMPARTMENTS reflect sampling scope and the corroboration requirement, respectively, rather than data quality.

## § 4 Limitations and next steps

This milestone produces a broad, traceable candidate list for downstream filtering, not a validated surfaceome. Four limitations matter for interpretation.

1. **DeepTMHMM coverage.** Predictions come from a pre-existing cohort of 2,360 proteins. Extending DeepTMHMM – alongside SignalP 6.0 and GPI-anchor predictors – to the full reviewed proteome is planned.
2. **Organelle-lumen exclusion.** A small number of proteins enter on UniProt evidence whose only subcellular-location support is an organelle lumen; these will be filtered out in a refinement pass.
3. **Tissue and cell-type filtering.** The full HPA data product – tissue specificity, single-cell, cancer atlas, Secretome and Predicted-membrane class – is not yet incorporated; a prerequisite for modality ranking.
4. **Ambiguous historical accessions.** 5 proteins whose historical accessions split into multiple current primaries are excluded from positive calls until manually reviewed.

# *§ 5* References

*First-party citations for each source's methodology and data, grouped by provider. Each entry includes direct view / PDF / download links and a one-line description of what the reference is cited for in §§ 2.1–2.7.*

## § 5.1 UNIPROT

[1] UniProt Consortium. *UniProt: the Universal Protein Knowledgebase in 2025.* Nucleic Acids Research 53(D1):D609–D617, 2025.
view · pdf · UP000005640
*Canonical description of UniProtKB architecture, the Swiss-Prot/TrEMBL review split, and the 2025 content update.* CITED § 2.1

[2] UniProt. *Subcellular location* (help page).
view · subcell.txt
*Controlled vocabulary used in SUBCELLULAR LOCATION comment lines; the basis for the surface-topology filter (cell surface, integral, GPI-anchored, transmembrane).* CITED § 2.1

[3] UniProt. *Release 2026_01.* Release notes, 2026-01-28.
release notes · downloads
*The exact snapshot used for accession-history reconciliation across all seven sources. Release cadence is ~8 weeks.* CITED § 2 INTRO

## § 5.2 GENE ONTOLOGY

[4] Gene Ontology Consortium. *Download Annotations.* Current release landing page; human GAF release 2026-03-25.
downloads · goa_human.gaf.gz
*The human GO annotation snapshot and its release cadence; source of the gene–term associations filtered to surface-relevant terms.* CITED § 2.2

[5] Gene Ontology Consortium. *GO Annotation File (GAF) format 2.2.*
spec
*Defines the single-(gene, term) association record carrying an evidence code, which we filter on to drop purely electronic annotations.* CITED § 2.2

[6] Gene Ontology Consortium. *Guide to GO evidence codes.*
view · GOA IEA pipeline
*Evidence-code taxonomy (Experimental, Phylogenetic, Computational Analysis, Author Statement, Curator Statement, Automatically-assigned); the basis for excluding IEA-only annotations from the positive call.* CITED § 2.2

## § 5.3 SURFY

[7] Bausch-Fluck, D., Goldmann, U., Müller, S., *et al. The in silico human surfaceome.* PNAS 115(46):E10988–E10997, 2018.
view · pdf · resource
*Random-forest classifier trained on CSPA-positive proteins; source of the 2,886-protein prediction set with 1% / 5% / 15% FPR tiers.* CITED § 2.3

# § 5 References *continued – CSPA, DeepTMHMM, HPA, COMPARTMENTS*


## § 5.4 CSPA

[8] Bausch-Fluck, D., Hofmann, A., Bock, T., *et al. A Mass Spectrometric-Derived Cell Surface Protein Atlas.* PLOS ONE 10(4):e0121314, 2015.
view · pdf · S2 (data)
*Cell Surface Capture glycoproteomics across 41 human and 31 mouse cell types; the validated-surfaceome table used for the CSPA-positive input and its high-confidence / putative / unspecific / blank categories.* CITED § 2.4

## § 5.5 DEEPTMHMM

[9] Hallgren, J., Tsirigos, K. D., Pedersen, M. D., *et al. DeepTMHMM predicts alpha and beta transmembrane proteins using deep neural networks.* bioRxiv 2022.04.08.487609, 2022.
view · pdf · service
*ESM-1b encoder plus CRF decoder producing the Alpha TM / SP+TM / Beta / SP-only / Globular class labels used as the topology-based surface call.* CITED § 2.5

## § 5.6 HUMAN PROTEIN ATLAS

[10] Thul, P. J., Åkesson, L., Wiking, M., *et al. A subcellular map of the human proteome.* Science 356:eaal3321, 2017.
view
*The original immunofluorescence-based subcellular atlas underlying the HPA subcellular data product.* CITED § 2.6

[11] Human Protein Atlas. *Subcellular localization method – imaging.* Method page.
view · license
*Current description of antibody validation, reliability scoring, and the knowledge-based comparison against UniProtKB/Swiss-Prot.* CITED § 2.6

[12] Human Protein Atlas. *Subcellular location data (v25.0, Ensembl 109).* Released 2025-11-11.
view · subcellular_location.tsv.zip · releases
*Reliability classes (Enhanced / Supported / Approved / Uncertain), named localizations, and the GO cellular-component identifier column used for the v25 HPA call.* CITED § 2.6

## § 5.7 JENSENLAB COMPARTMENTS

[13] Binder, J. X., Pletscher-Frankild, S., Tsafou, K., *et al. COMPARTMENTS: unification and visualization of protein subcellular localization evidence.* Database (Oxford) 2014:bau012, 2014.
view · pdf
*Defines the knowledge / experiments / text-mining / predictions four-channel integration and the five-star scoring used for the ≥ 3-star gate.* CITED § 2.7

[14] JensenLab. *COMPARTMENTS – Downloads.* Per-channel TSVs, weekly updated.
downloads · textmining.tsv · experiments.tsv
*Per-channel files used to reconstruct the corroboration-gated call (text-mining + non-HPA portion of experiments), holding UniProt / GO / HPA duplicates out of the COMPARTMENTS evidence.* CITED § 2.7